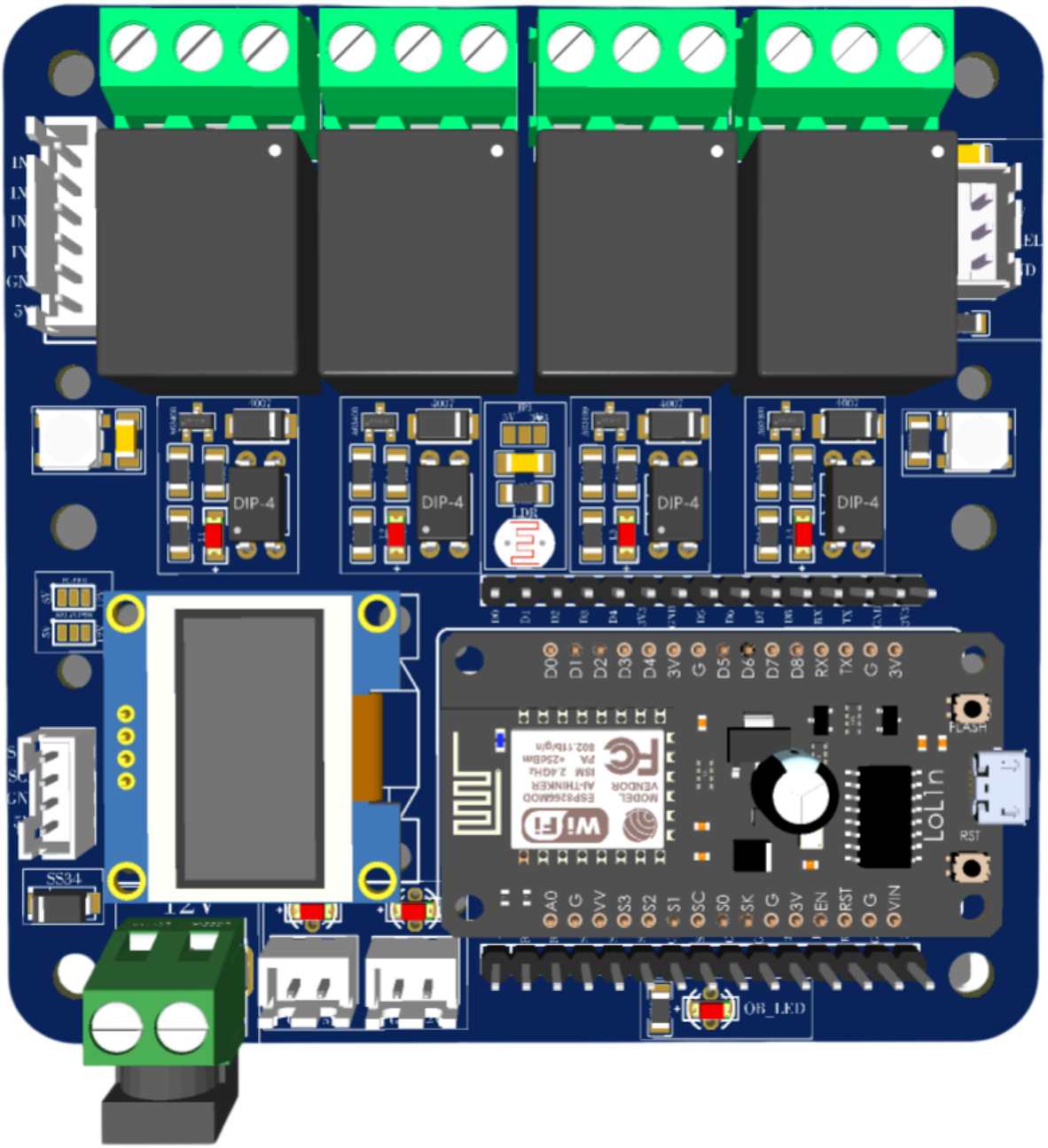


DEVSKY's 4 CH RELAY PCB

VERSION : 11.0.1

PRESENTED BY:

DEVSKY AND TECHNICAL

Overview

The DEVSKY ESP8266 4-Channel Relay Controller is an industrial-grade WiFi automation board designed for switching AC loads using the ESP8266 microcontroller. The board integrates opto-isolated relay drivers, 12V power input, and a 0.96” OLED display for real-time monitoring and control applications.

KEY FEATURES

- ESP8266 WiFi Module
- 4 × 10A Relay Outputs
- 12V DC Input
- Onboard 7805 Voltage Regulator
- 0.96” OLED Display Support
- Opto-isolated Relay Driver Stage
- Screw Terminal Load Connections
- USB Programming Interface
- LDR Analog Light Sensing
- Jumper Selectable Voltage Configuration

PIN CONFIGURATION

GPIO	FUNCTION
D3	RELAY 1
D4	RELAY 2
D7	RELAY 3
D8	RELAY 4
D1	OLED SCL
D2	OLD SDA
AO	LDR INPUT
DO	ONBOARD LED

POWER SPECIFICATIONS

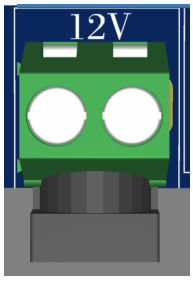
🔹 Power Supply Configuration

The DEVSKY 4-Channel Relay Controller operates from a **12V DC input supply**.

• 12V INPUT

• The board accepts 12V DC through the main power terminal.
• Two input connectors are provided:

- DC Barrel Jack
- Screw Terminal Connector



Users may use either connector as per installation requirement.

The 12V line is used to power:

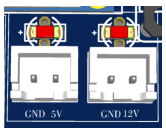
• Relay coils (when 12V selected)
• PCF module (if 12V selected via jumper)

• 5V REGULATION (USING 7805)

- An onboard 7805 linear voltage regulator steps down 12V to a stable 5V output.
- The regulated 5V is used for:
- • Logic circuitry
  • OLED display
  • Optional 5V relay operation
  • PCF module (if 5V selected via jumper).

• POWER OUTPUT CONNECTOR (WHITE CONNECTOR)

The white connector provides:



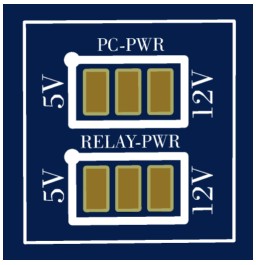
• 12V Output
• Regulated 5V Output

This allows external modules and peripherals to be powered directly from the board.

• RELAY & PCF POWER SELECTION (JUMPER CONTROLLED)

The board includes jumper-selectable voltage routing.

Relay Supply Options:



• 12V (for 12V relay coils)
• 5V (for 5V relay coils)

PCF Supply Options:

• 12V
• 5V

This design allows flexible relay configuration without PCB modification.

• ESP8266 OPERATING VOLTAGE

• The ESP8266 operates at 3.3V logic level.
• 3.3V is provided by the onboard regulator of the ESP8266 development board.

⚠ ESP8266 GPIO pins are NOT 5V tolerant.

• OLED DISPLAY SUPPLY

• Operates at 5V supply
• Communicates via I2C interface

The 5V is sourced from the regulated 7805 output.

Note: A separate I2C header is available for additional external devices.

• CONTROL HEADER (J1)

The J1 header provides relay control signal inputs and logic power.

Pin Configuration:

• IN1 – Relay Channel 1
• IN2 – Relay Channel 2
• IN3 – Relay Channel 3

• IN4 – Relay Channel 4 • GND – Common Ground
• 5V – Logic Supply

This header allows external microcontrollers to control the relay channels directly.

⚠ Ensure logic compatibility before connection.

• LDR INPUT (A0)

An onboard **LDR** is connected to the **A0 analog pin** for light sensing.

Jumper Selectable Supply:

• 5V mode
• 3.3V mode

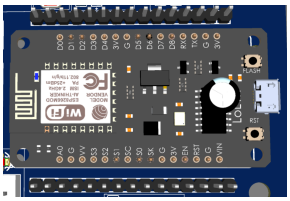
The jumper allows safe operation with both 5V and 3.3V systems.

⚠ Ensure correct voltage selection before use.

• ESP8266 EXPANSION HEADER

Separate male headers are provided on both sides of the ESP8266 module.

These expose:

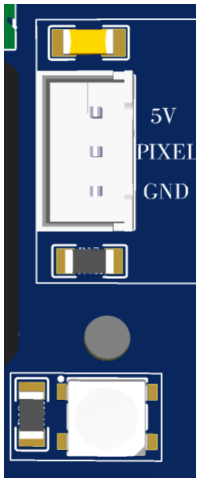


• GPIO pins
• Power pins
• I2C / SPI / UART interfaces

This enables easy external module integration.

⚠ All external connections must follow 3.3V logic compatibility.

• RGB Pixel (Left & Right)



Dedicated Pixel are provided on both the **Left (L)** and **Right (R)** sides of the PCB for external RGB pixel modules.

Pin Configuration:

• 5V – Pixel supply voltage
• DATA – Signal input/output (D5 of ESP8266)
• GND – Ground reference

These connectors allow easy integration of addressable LED pixels (e.g., WS2812 type) for visual indication or decorative lighting.

The dual-side placement enables flexible installation and daisy-chaining of pixel modules.

⚠ Ensure correct polarity before connection.
⚠ Use appropriate current-rated power supply when driving multiple pixels.

Conclusion

The DEVSKY 4-Channel Relay Controller is a flexible and expandable automation platform built around the ESP8266.

With:

• Separate logic and relay power inputs
• Jumper-selectable voltage configuration
• Onboard 7805 regulation
• Expandable GPIO access
• Integrated OLED support
• LDR sensing capability

The board is suitable for IoT, home automation, and industrial control applications.

The modular power architecture ensures stable performance while maintaining compatibility with both 5V and 3.3V systems.

Support & Contact

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Disclaimer

This product is intended for controlled environments.
Proper insulation and safety precautions must be followed when switching high-voltage loads.

DevSky Technical is not responsible for damage caused by improper installation or misuse.